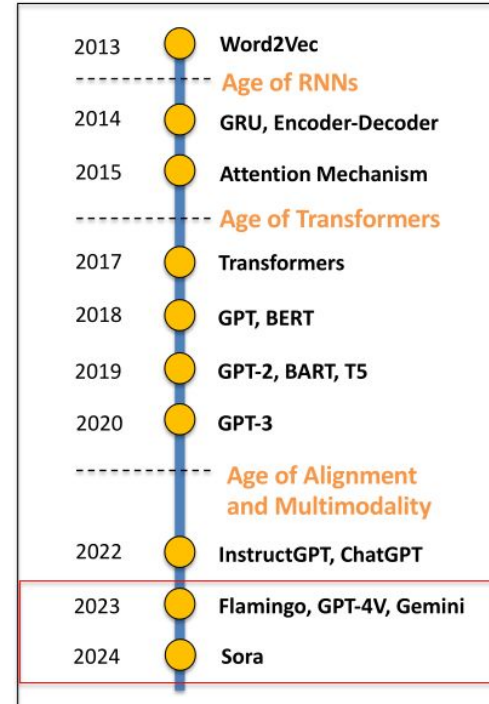
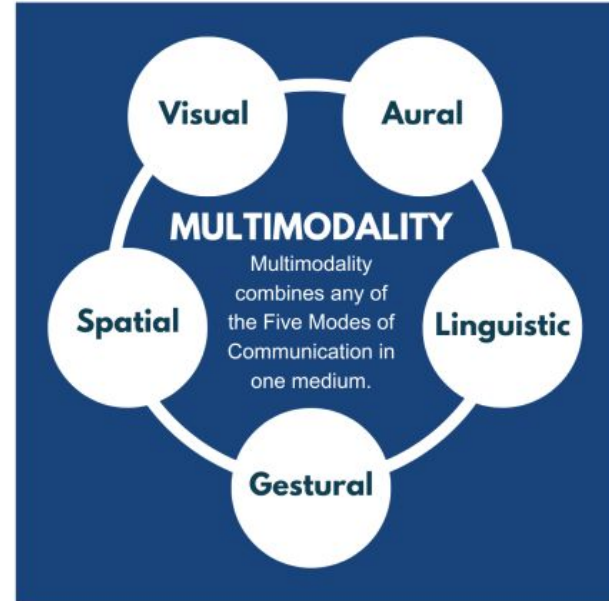


From Word2Vec to GPT (10 Years of Deep NLP)

1. Tokenization and Word2Vec
2. RNNs, LSTM, GRU, Encoder-Decoder and Attention Mechanism
3. Self-Attention and Transformers
4. LLMs: BERT, GPT, BART, T5
5. Alignment: InstructGPT, ChatGPT
6. **Multimodality: GPT-4V, Gemini, Sora**

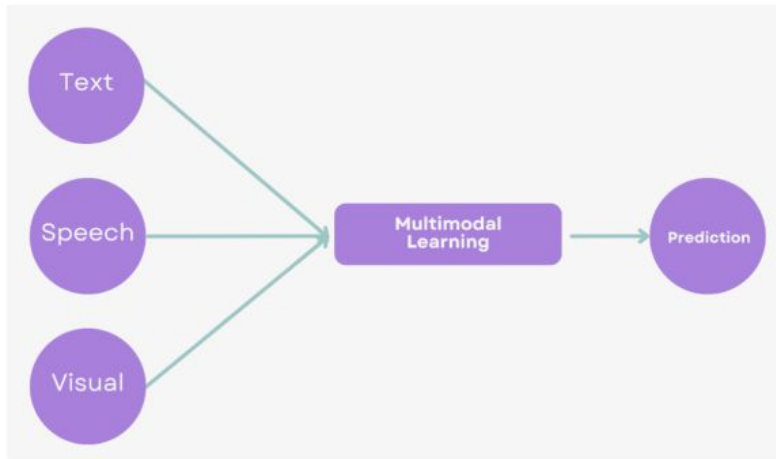


- **What and why is multimodality?**
- **Contrastive Techniques**
 - CLIP (2020), ImageBind (2021)
- **Masked Visual Language Models**
 - VisualBERT (2019) and ViLBERT (2019)
- **Unified Multimodal Models**
 - BLIP (2022), VL-T5 (2022)
- **Generative Multimodal Models**
 - Frozen (2022), Flamingo (2023), LLaVA (2023), GPT-4V (2023), Gemini (2023)



What is Multimodality?

- **Multimodality** refers to the integration of multiple modes or modalities in a given context.
- When applied in the realm of AI, it commonly refers to **combining text with one or more other modalities** such as images, speech, audio, or others.

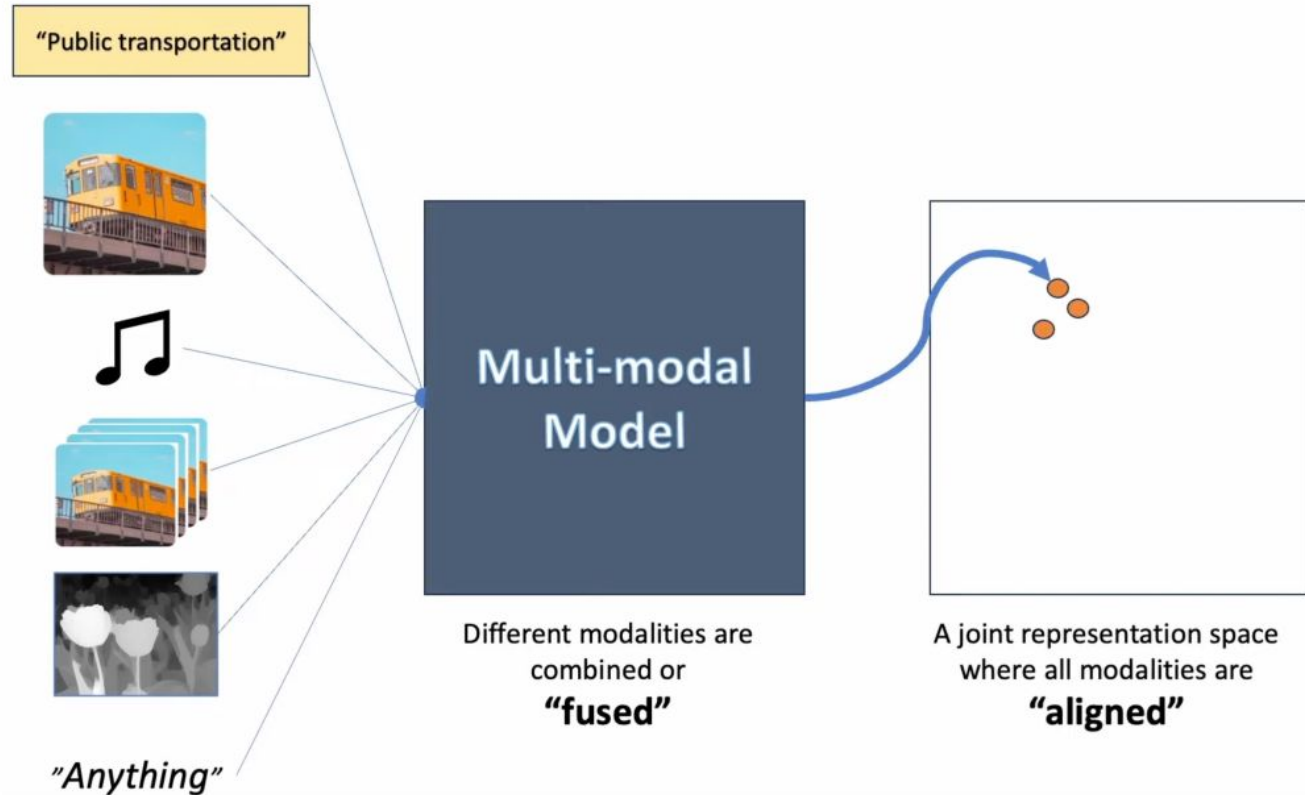


Multimodal Learning
combines data from
multiple modalities
to train machine
learning models

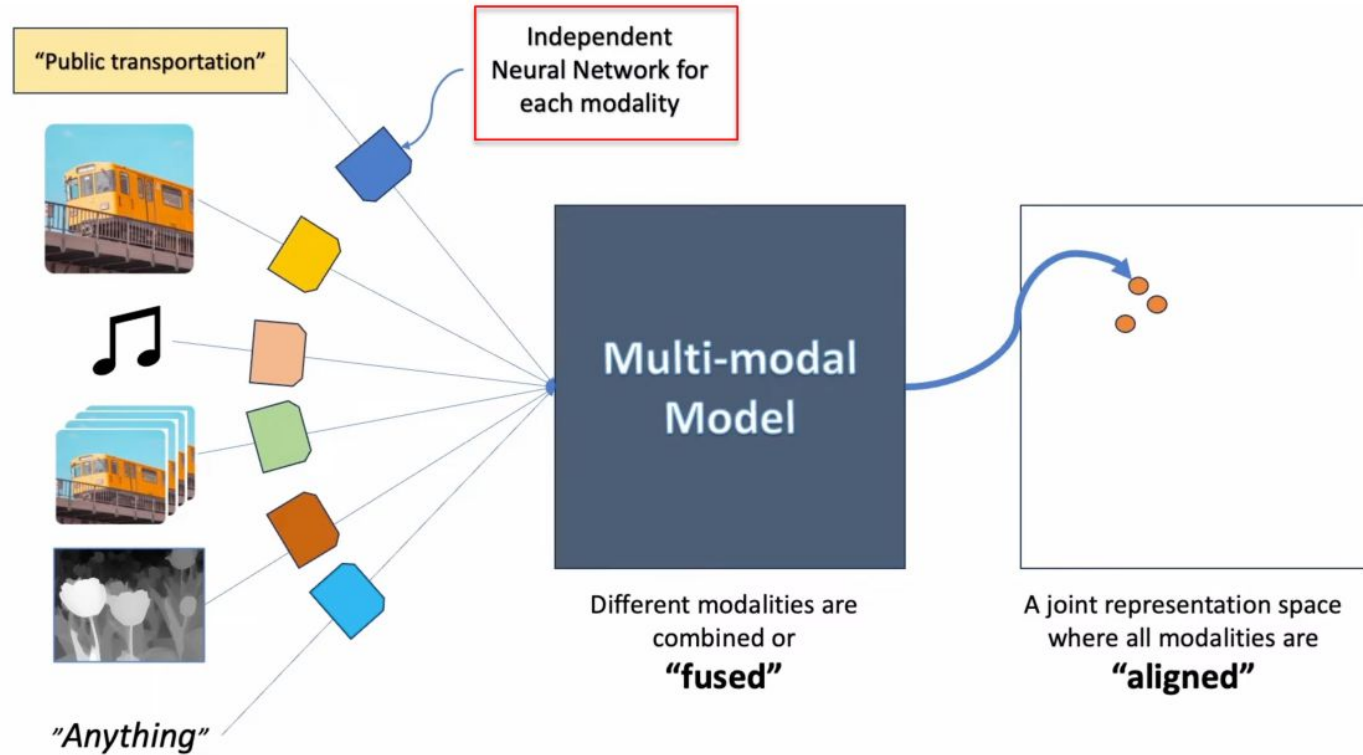
Why does Multimodality Matter?

- Multimodality matters for several reasons:
 - **Faithfulness**: Human experiences are inherently multimodal, involving multiple senses and modalities.
 - **Practicality**: The internet and many applications we use daily incorporate multiple modalities, making multimodality essential for understanding and processing information.
 - **Data efficiency and availability**: Multimodal data provides rich and high-bandwidth information compared to language alone. This richness makes it conducive for better learning and understanding.
 - **Scaling**: As the need for more data arises, high-quality text data becomes limited. Multimodal data allows for a broader range of information sources, expanding the available data for training models.
- Multimodality is at the forefront of the new revolution in foundation models:
 - **Large Language Models (LLMs) => Large Multimodal Models (LMMs)**

Multimodal Learning Architecture

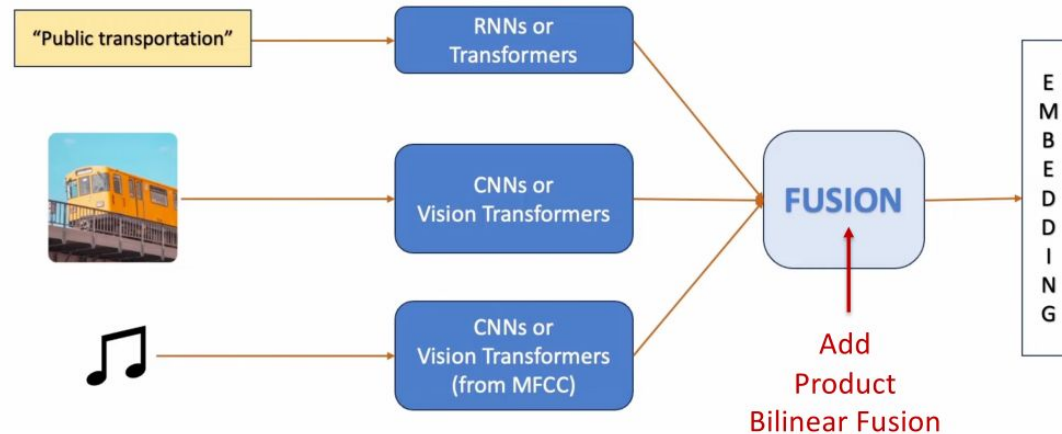


Multimodal Learning Architecture



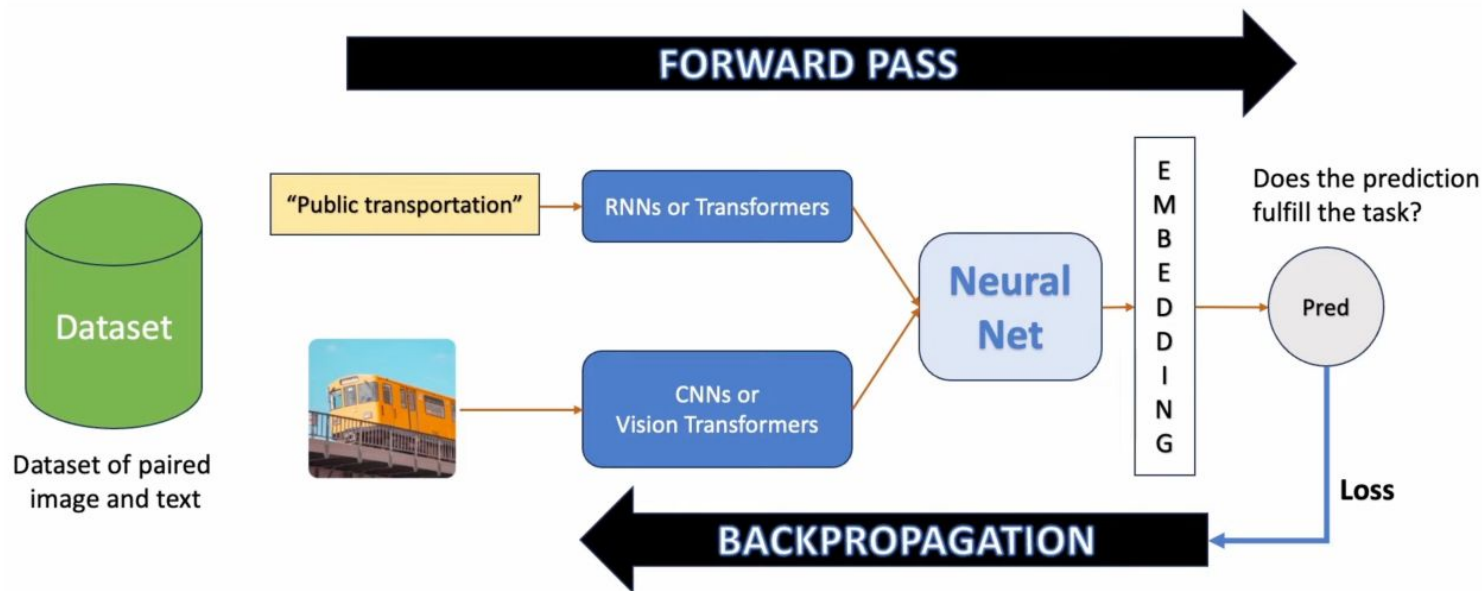
Multimodal Learning Architecture

- **Step 1: Unimodal Embeddings** that process each input independently with an input-appropriate neural network
- **Step 2: Fusion of embeddings** that combine each unimodal embedding into one multimodal representation



Alignment of Embeddings

- Train the network on a predefined task to align the embeddings.



Contrastive Learning

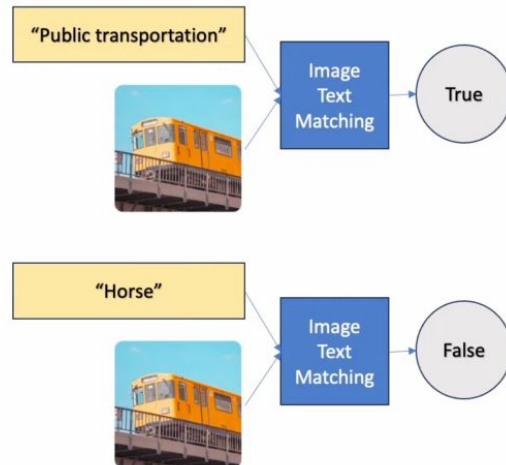
(OpenAI **CLIP** and Meta **ImageBind**)

Contrastive Learning

- **Contrastive Learning** utilize **discriminative training** to enable models to differentiate or identify similarities across different modalities present in the input data.
- Discriminative training focuses on classifying or determining boundaries within a predefined set of labels, rather than generating new content.
- Well-known multimodality contrastive techniques:
 - CLIP
 - ImageBind

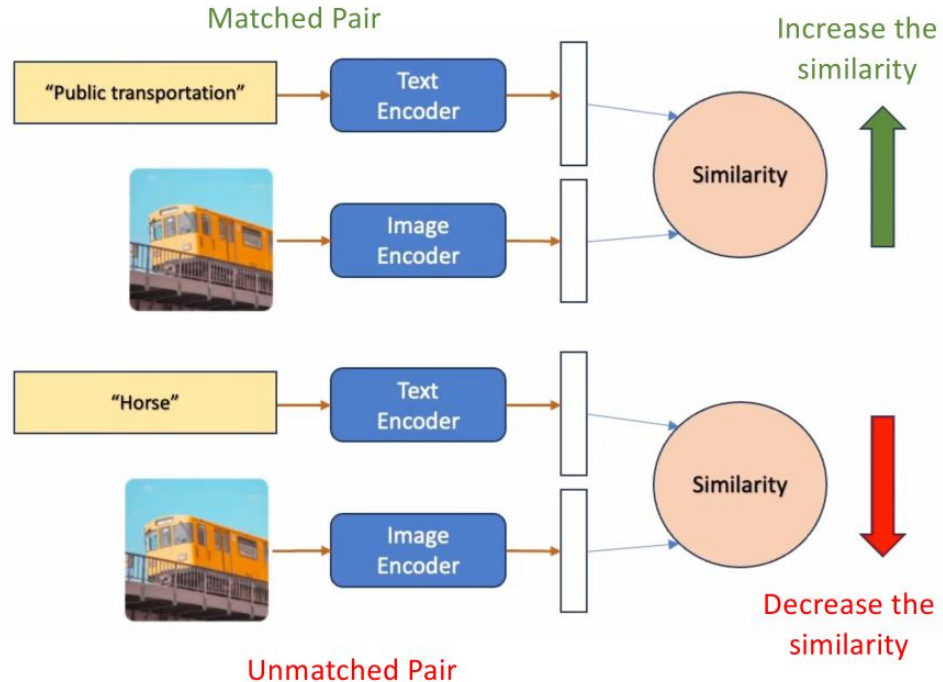
Discriminative Training

Image – Text Matching



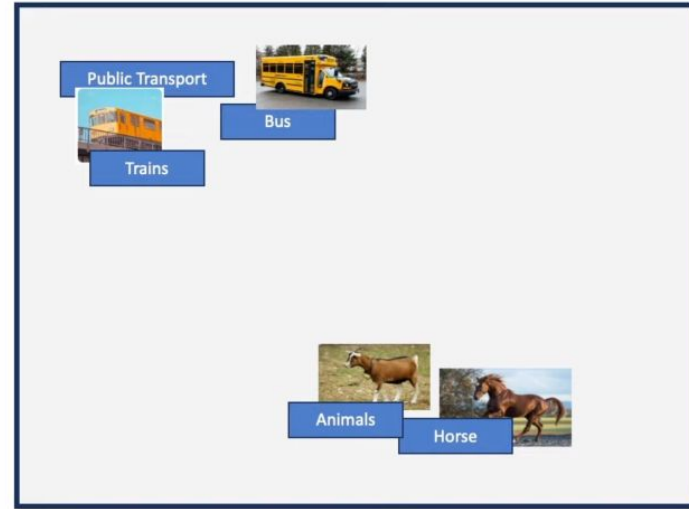
Contrastive Learning

- Increases the similarity (dot-product) of embeddings between matched input pairs
- Decrease the similarity (dot-product) of unmatched pairs



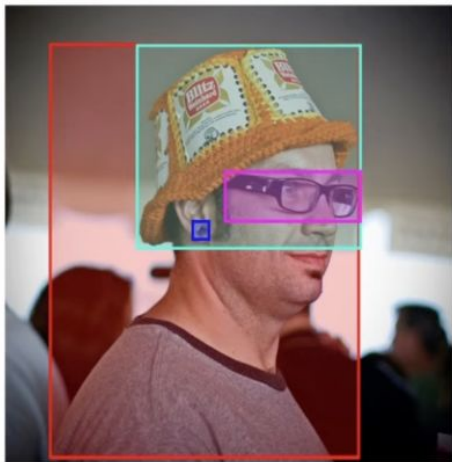
Contrastive Learning

- **Increases the similarity (dot-product)** of embeddings between matched input pairs between matched input pairs
- **Decrease the similarity (dot-product)** of unmatched pairs
- A **joint embedding space** where the embeddings of **paired text and image are close to each other**, while being further away from dissimilar ones



Joint embedding space for images and captions

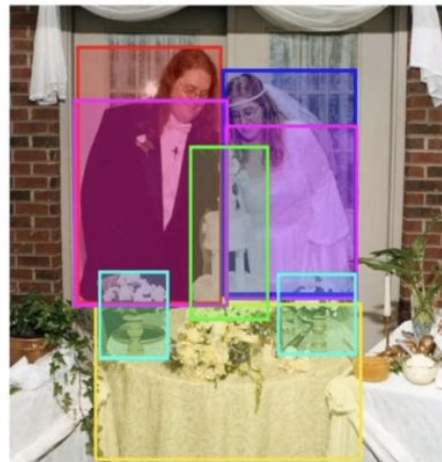
Dataset: Image – Text Pairs



A man with pierced ears is wearing glasses and an orange hat.
A man with glasses is wearing a beer can crocheted hat.
A man with gauges and glasses is wearing a Blitz hat.
A man in an orange hat starring at something.
A man wears an orange hat and glasses.

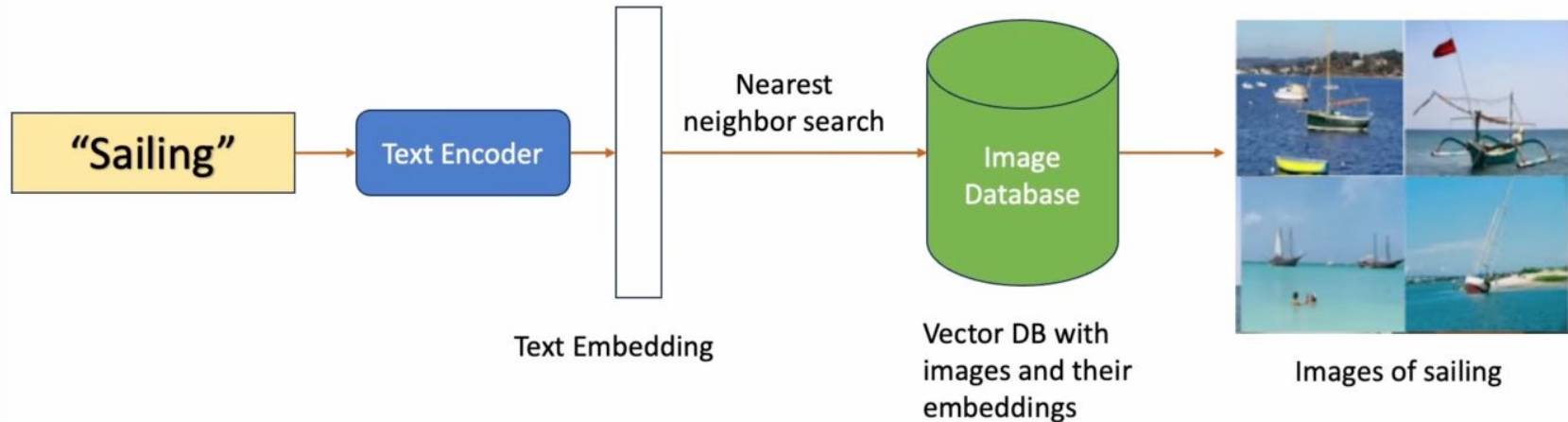


During a gay pride parade in an Asian city, some people hold up rainbow flags to show their support.
A group of youths march down a street waving flags showing a color spectrum.
Oriental people with rainbow flags walking down a city street.
A group of people walk down a street waving rainbow flags.
People are outside waving flags .



A couple in their wedding attire stand behind a table with a wedding cake and flowers.
A bride and groom are standing in front of their wedding cake at their reception.
A bride and groom smile as they view their wedding cake at a reception.
A couple stands behind their wedding cake.
Man and woman cutting wedding cake.

- Search images from a database relevant to the text query “Sailing”.



CLIP (2021-03)

Contrastive Language-Image Pretraining

Connecting Text and Image

Radford, Alec, et al. "Learning transferable visual models from natural language supervision," ICML 2021

- **CLIP (Contrastive Language-Image Pre-training)** is an influential **multimodal model** that established a new state-of-the-art for **connecting image and language**.

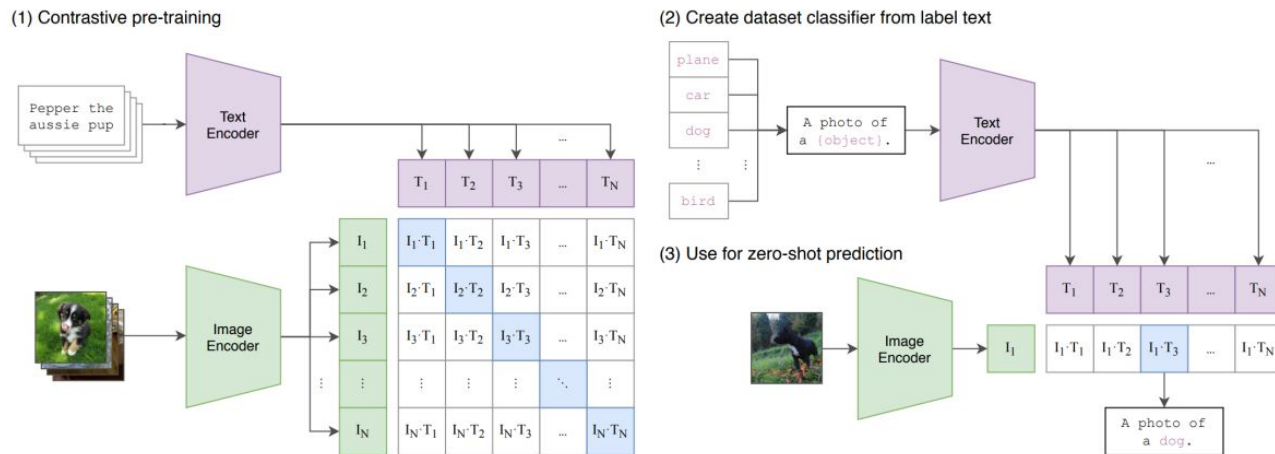
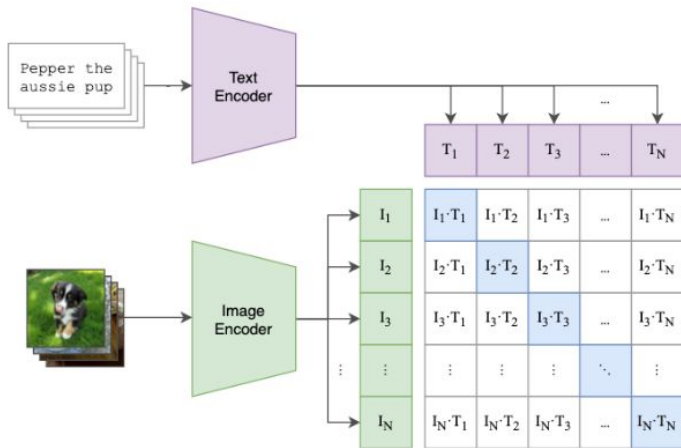


Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset's classes. <https://arxiv.org/pdf/2103.00020v1.pdf>

- **CLIP's contrastive pre-training** can train models that can **understand and match Image-Text Pairs**

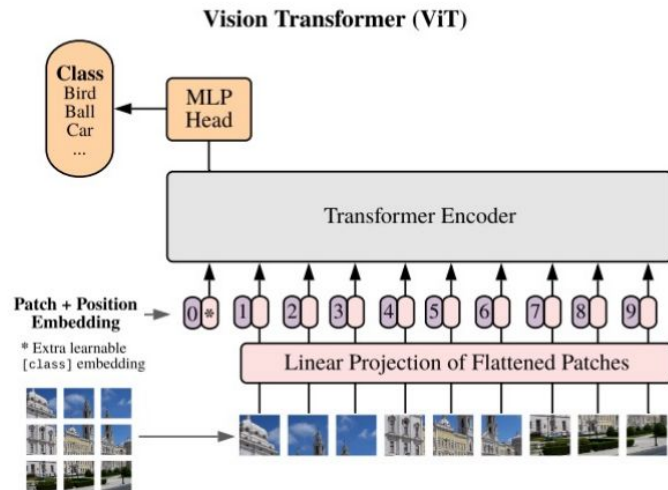
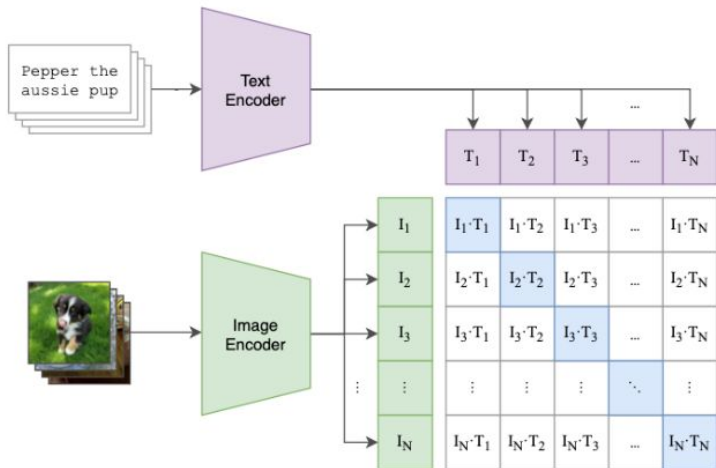
(1) Contrastive pre-training



- **Dataset: WebImageText (WIT)** is a newly constructed dataset of **400 million (image, text) pairs** on the Internet
- **CLIP's Text Encoder** is a 12-layer transformer based on GPT-2 **to extract semantic features from text.**
- **CLIP's Image Encoder** can be a CNN-based ResNet or a Vision Transformer **to extract semantic visual features from images.**
- They are **projected to match the dimensionality of the text and image embeddings**

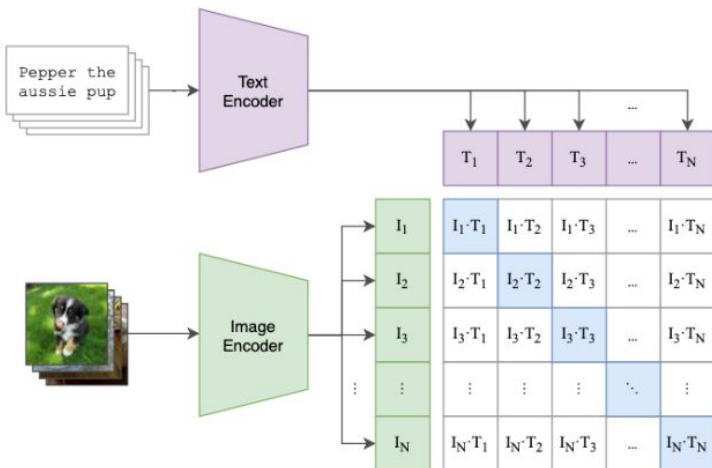
- CLIP's Image Encoder can be a CNN-based ResNet or a Vision Transformer (ViT) to extract semantic visual features from images.

(1) Contrastive pre-training



- **WebImageText (WIT) Dataset:** CLIP was trained using a freshly assembled dataset comprising **400 million pairs of images and corresponding text** sourced from the Internet.

(1) Contrastive pre-training



- Batch size = 32,768

- Cosine distance

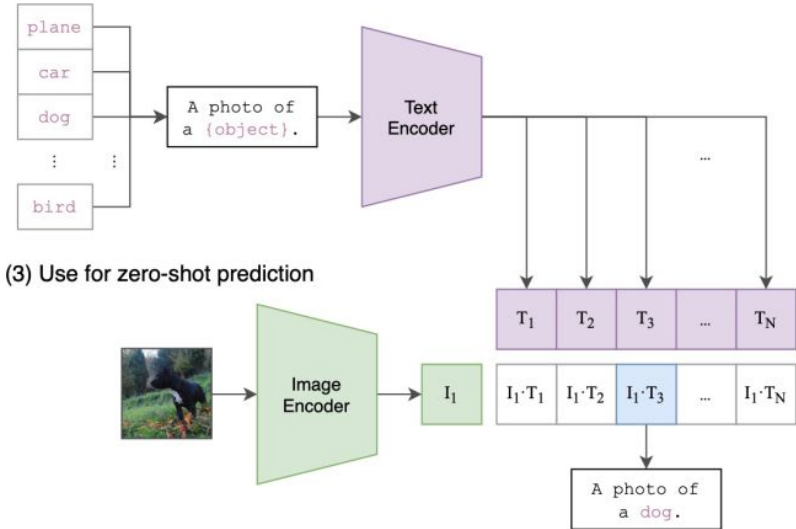
$$\min \left(\sum_{i=1}^N \sum_{j=1}^N (I_i \cdot T_j)_{i \neq j} - \sum_{i=1}^N (I_i \cdot T_i) \right)$$

Why is CLIP a Game Changer?

- CLIP enabled **Zero-Shot Classification** without Finetuning

- Because zero-shot classification allows you to work on any data and still get good results.
- Unlike conventional techniques where the model had to be trained specifically to recognize some objects, CLIP can identify many things without finetuning.

(2) Create dataset classifier from label text



Why is CLIP a Game Changer?

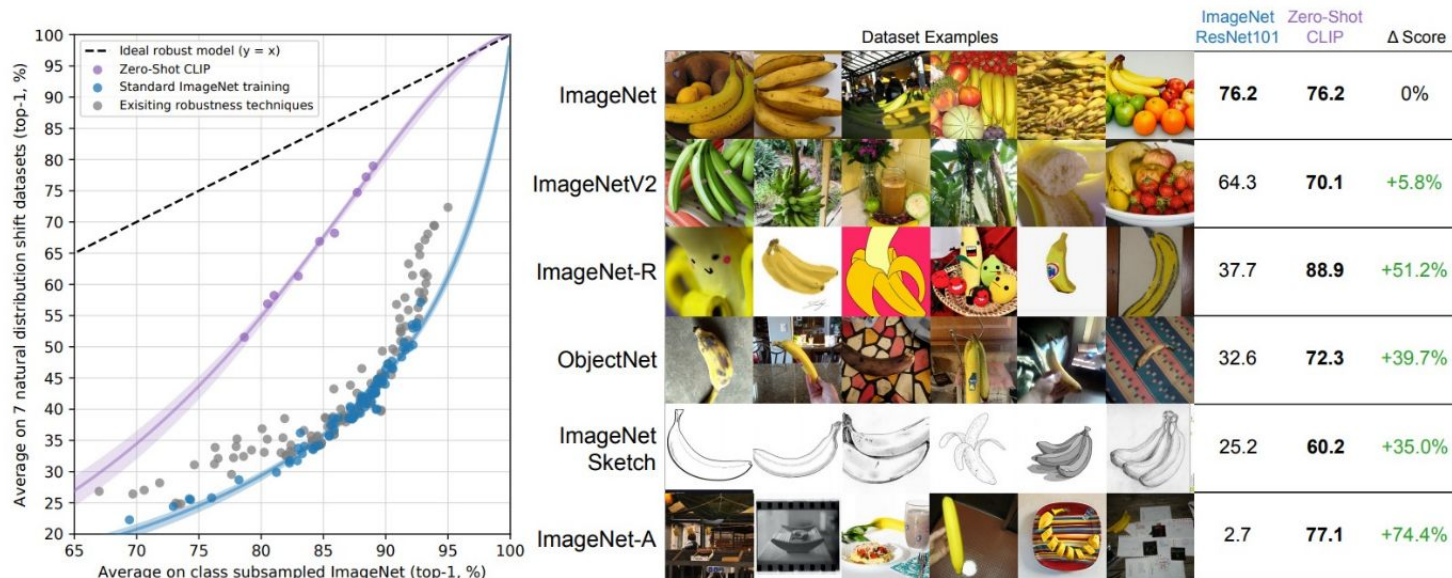


Figure 13. Zero-shot CLIP is much more robust to distribution shift than standard ImageNet models. (Left) An ideal robust model (dashed line) performs equally well on the ImageNet distribution and on other natural image distributions. Zero-shot CLIP models shrink this “robustness gap” by up to 75%. Linear fits on logit transformed values are shown with bootstrap estimated 95% confidence intervals. (Right) Visualizing distribution shift for bananas, a class shared across 5 of the 7 natural distribution shift datasets. The performance of the best zero-shot CLIP model, ViT-L/14@336px, is compared with a model that has the same performance on the ImageNet validation set, ResNet-101.

Meta's ImageBind (2023)

ImageBind: a new way to 'link' AI across the senses

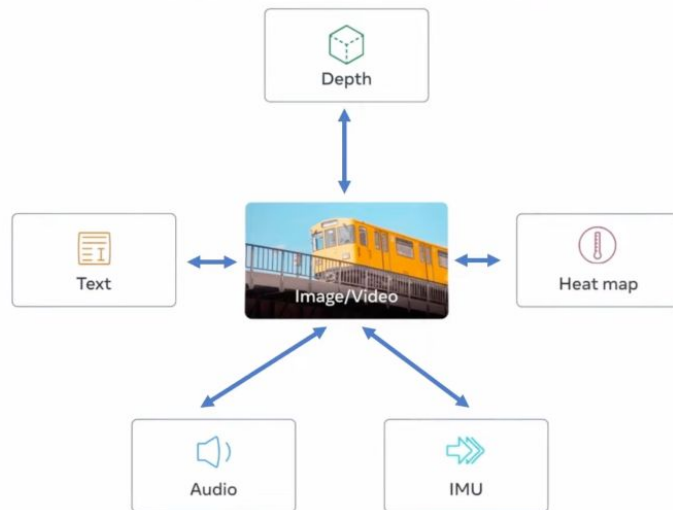
Introducing ImageBind, the first AI model capable of binding data from six modalities at once, without the need for explicit supervision. By recognizing the relationships between these modalities — images and video, audio, text, depth, thermal and inertial measurement units (IMUs) — this breakthrough helps advance AI by enabling machines to better analyze many different forms of information, together.

Explore the demo to see ImageBind's capabilities across image, audio and text modalities.

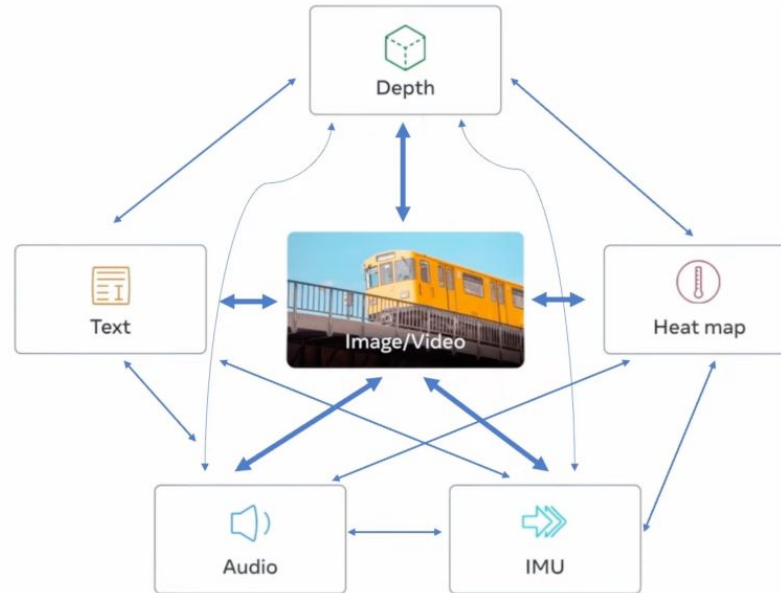
IMAGEBIND: One Embedding Space To Bind Them All

Rohit Girdhar* Alaaeldin El-Nouby* Zhuang Liu Mannat Singh
 Kalyan Vasudev Alwala Armand Joulin Ishan Misra*
 FAIR, Meta AI

<https://facebookresearch.github.io/ImageBind>



ImageBind can bind six modalities at the same time



Meta's ImageBind (2023)

1) Cross-Modal Retrieval

Audio	Images & Videos	Depth	Text
 Crackle of a Fire			"A fire crackles while a pan of food is frying on the fire." "Fire is crackling then wind starts blowing." "Firewood crackles then music..."
 Baby Cooing			"A baby is crying while a toddler is laughing." "A baby is laughing while an adult is laughing." "A baby laughs and something..."

2) Embedding-Space Arithmetic

 (+)  (→)   

3) Audio to Image Generation

 Dog  Engine  Fire  Rain

A dog barking

▶ 0:00 / 0:04  

 (+)  (→)    

Chirping birds

 (+)  (→)   

Claps

 (+)  (→)    

Church Bells

 (+)  (→)    

Thunderstorm



Masked Visual Language Modelling

(Multimodal Encoder-only Transformer Models)

VisualBert (2019) and ViLBERT (2019)

VisualBERT (2019-08)

- Splitting Images into regions

VISUALBERT: A SIMPLE AND PERFORMANT
BASELINE FOR VISION AND LANGUAGE

Liunian Harold Li[†], Mark Yatskar^{*}, Da Yin[°], Cho-Jui Hsieh[†] & Kai-Wei Chang[†]

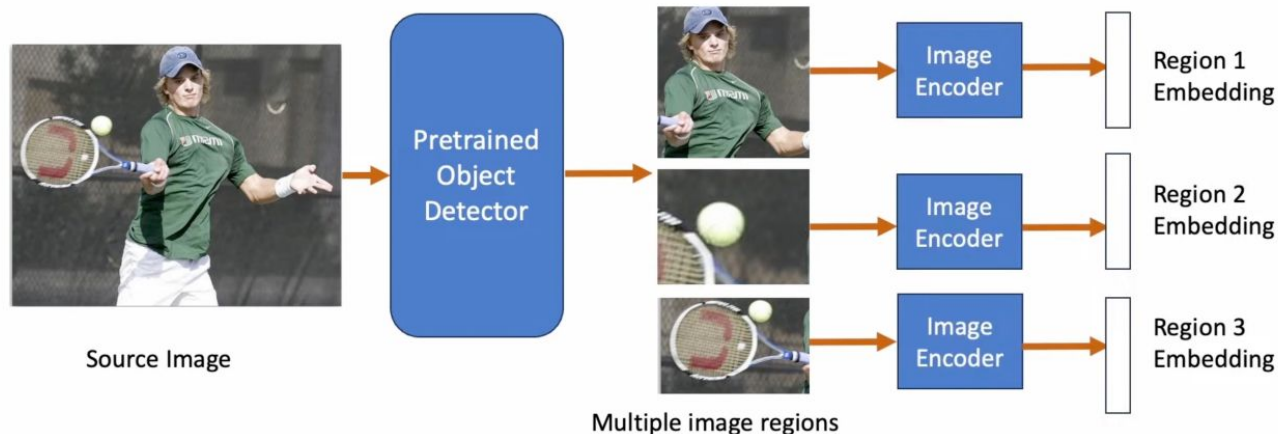
[†]University of California, Los Angeles

^{*}Allen Institute for Artificial Intelligence

[°]Peking University

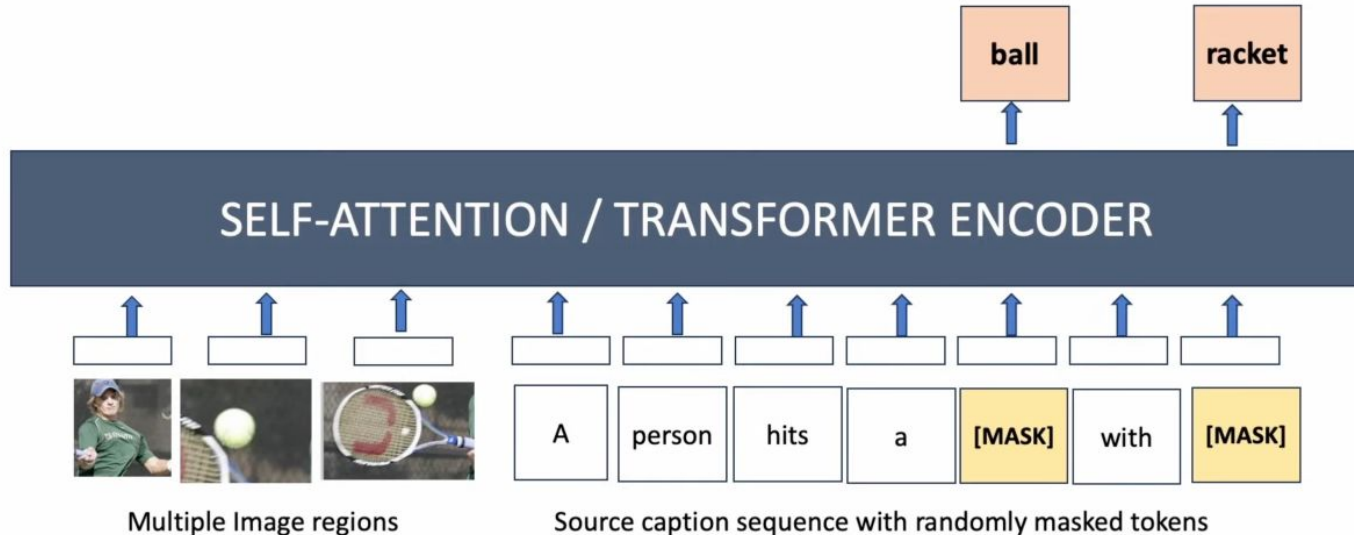
liunian.harold.li@cs.ucla.edu, marky@allenai.org,

wade.yin9712@pku.edu.cn, {chohsieh, kwchang}@cs.ucla.edu

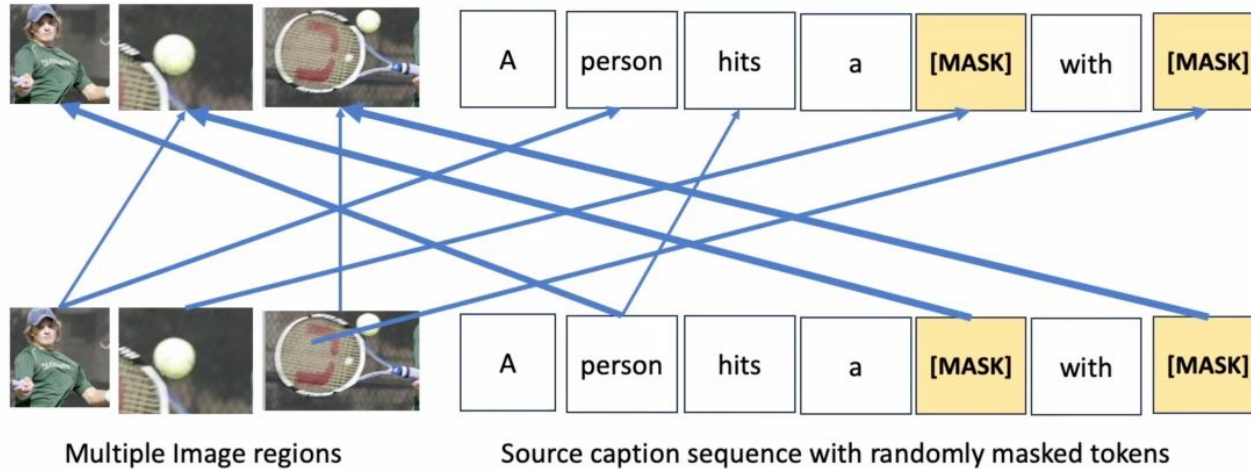


Source Caption:
A person hits a ball with a racket

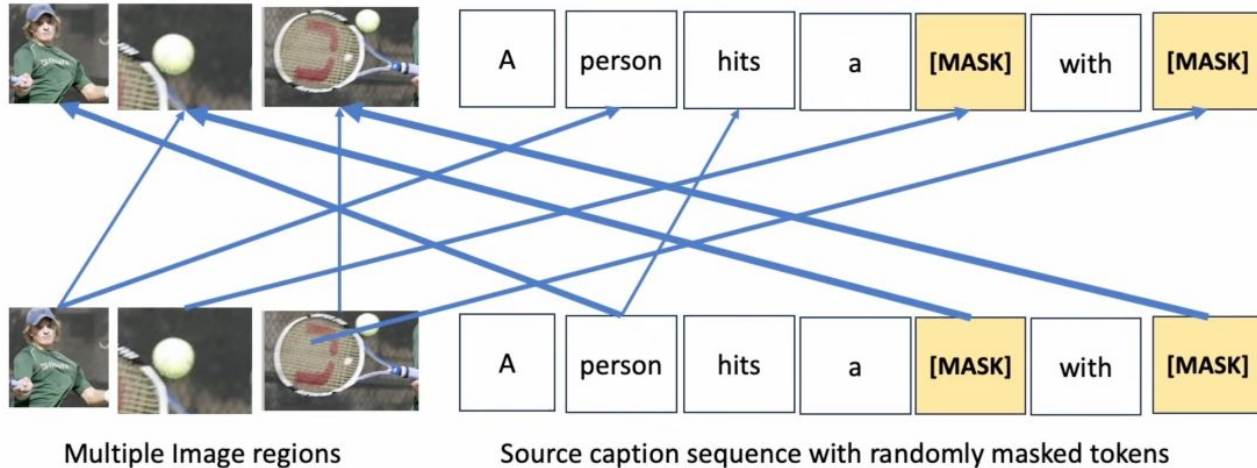
- **VisualBERT** ([Li et al. 2019](#)) feeds both text inputs and image regions into [BERT](#) such that it is able to discover the internal alignment between images and text with self-attention mechanism.



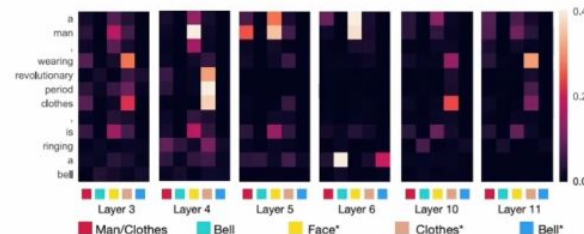
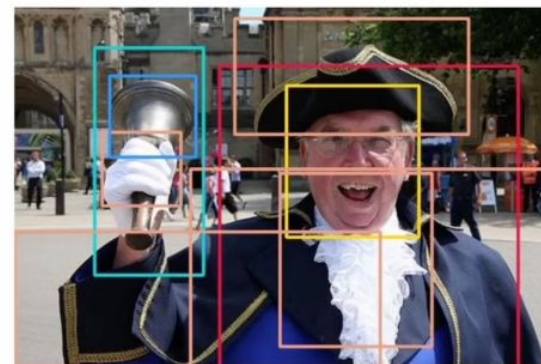
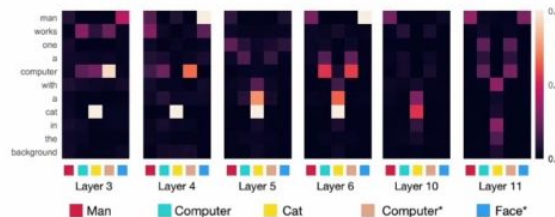
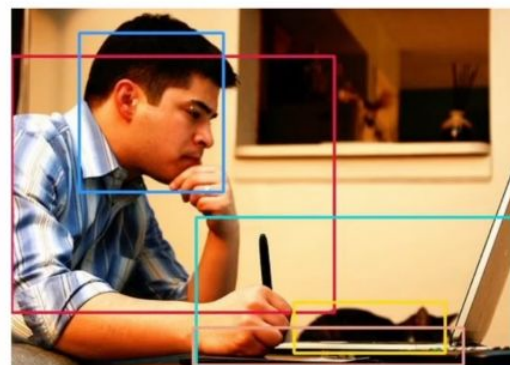
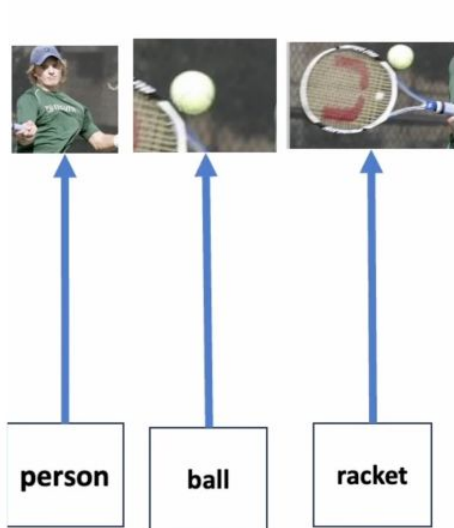
- At each self-attention transformer layer, we learn which token should “**attend to**” which other tokens of the input



- At each self-attention transformer layer, we learn which token should “**attend to**” which other tokens of the input



- Specific words learn to provide attention to specific regions of the image



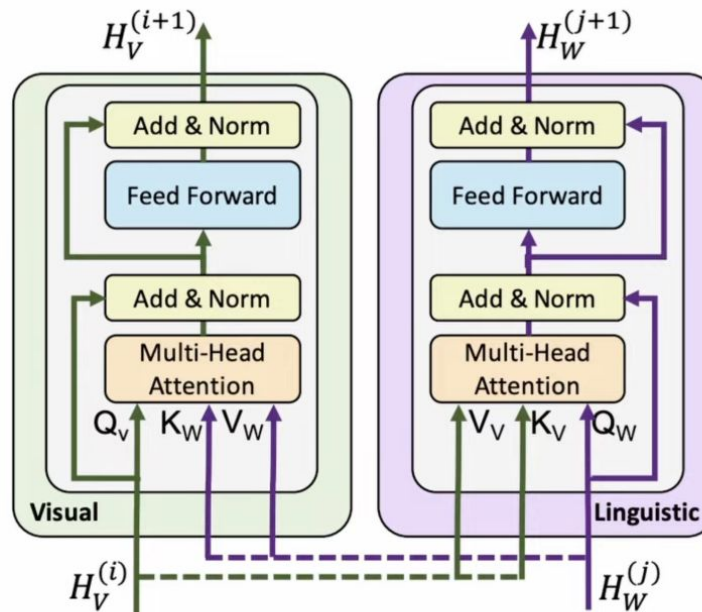
ViLBERT (2019)

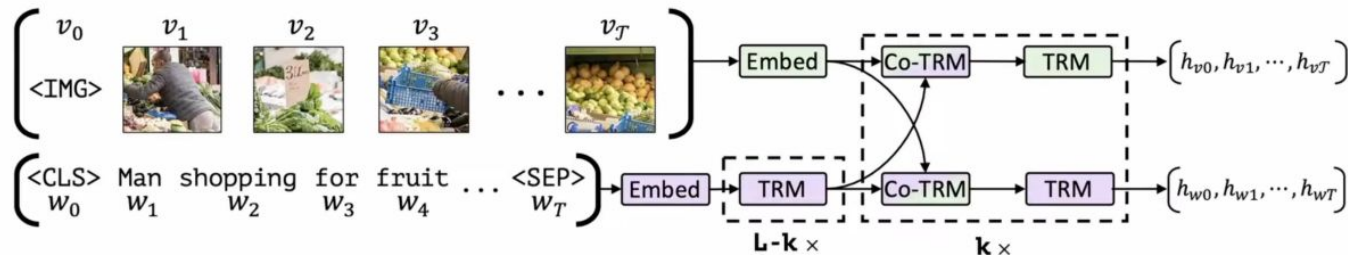
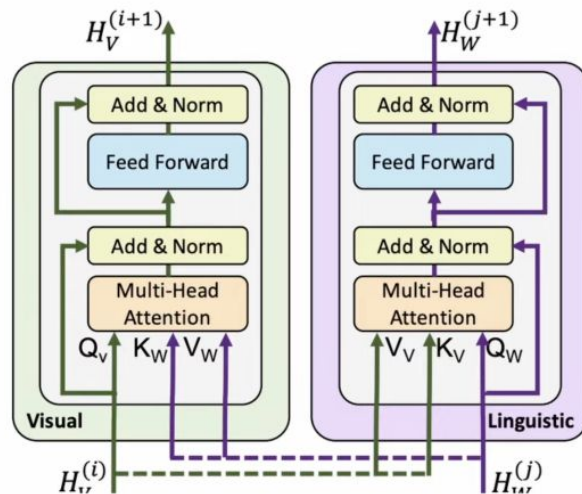
ViLBERT: Pretraining Task-Agnostic Visiolinguistic Representations for Vision-and-Language Tasks



Jiasen Lu¹, Dhruv Batra^{1,2}, Devi Parikh^{1,2}, Stefan Lee^{1,3}

¹Georgia Institute of Technology, ²Facebook AI Research, ³Oregon State University





Finetune to Downstream Tasks



The concept comes to life with a massive display of fireworks that will fill the grounds.



Happy young successful business woman in all black suit smiling at camera in the modern office.



A grey textured map with a flag of country inside isolated on white background .



New apartment buildings on the waterfront, in a residential development built for cleaner housing.



Is there something to cut the vegetables with?

VQA



Why is [person4] pointing at [person1]?

all he is telling [person4] that [person3] ordered the pancakes.
 all he just told a joke.
 all he is feeling accusatory towards [person1].
 all he is giving [person1] directions.

VCR Q→A

Rationale: a) is correct because...

a. [person4] has the pancakes in front of him.
 b. [person4] is taking everyone's order and asked for clarification.
 c. [person4] is looking at the pancakes both she and [person3] are smiling widely.
 d. [person4] is delivering food to the table, and she might not know whose order is whose.

VCR QA→R



Guy in yellow dribbling ball

Referring Expressions



Caption-Based Image Retrieval

Unified Modelling

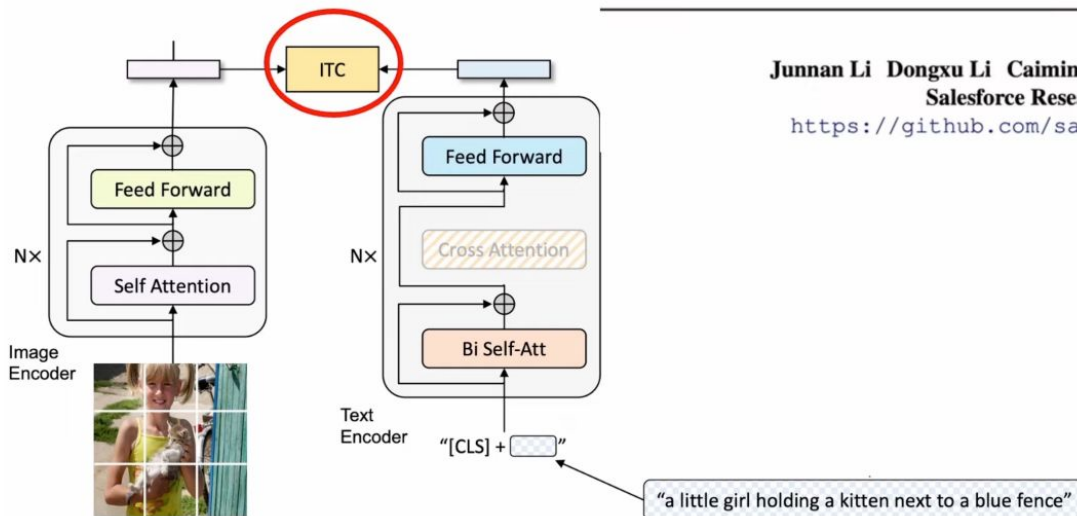
(Multimodal Encoder-Decoder Transformer Models)

BLIP (2022) and VL-T5 (2022)

- Trains jointly on multiple tasks and multiple datasets
- For example: one single model for question-answering, image-captioning, contrastive matching.
- **Benefits:**
 - Can be used for multiple tasks
 - General multipurpose embeddings
 - Combines many datasets, allowing larger training set.

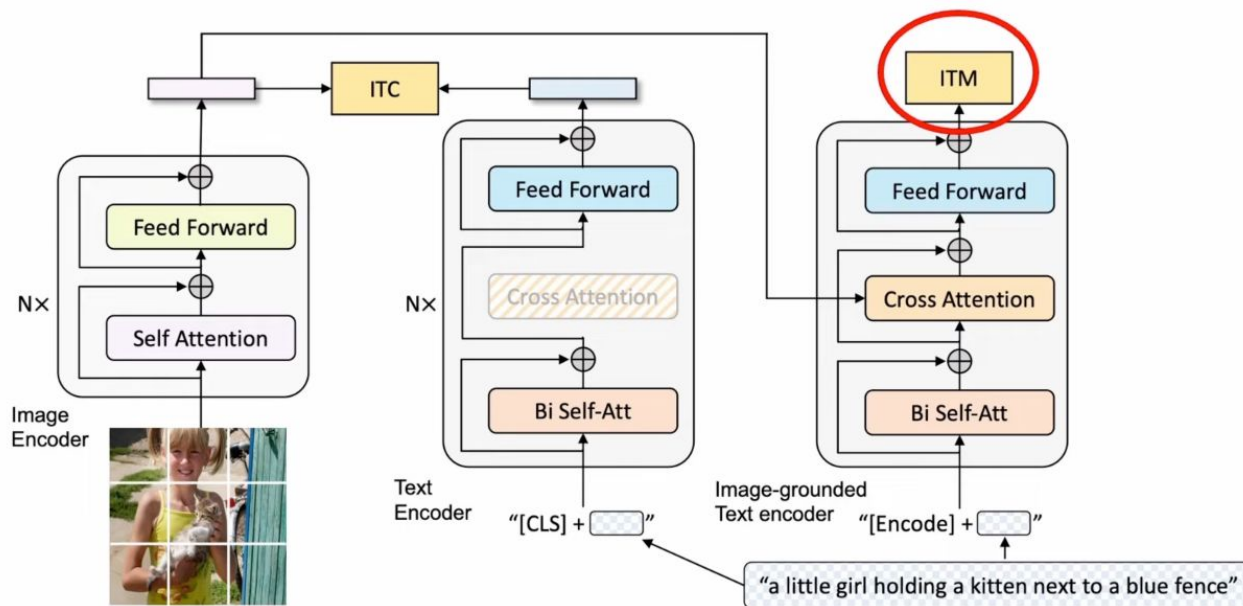
BLIP (2022)

BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation

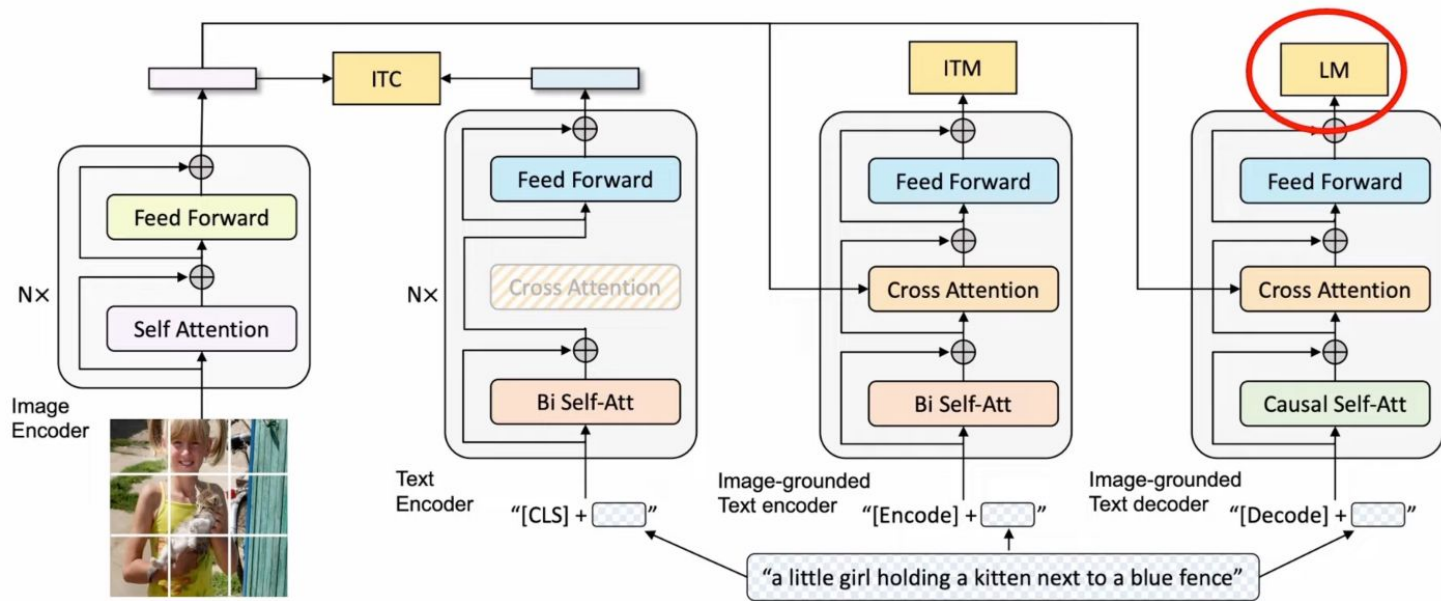


Junnan Li Dongxu Li Caiming Xiong Steven Hoi
Salesforce Research
<https://github.com/salesforce/BLIP>

Layers with same color share weights!



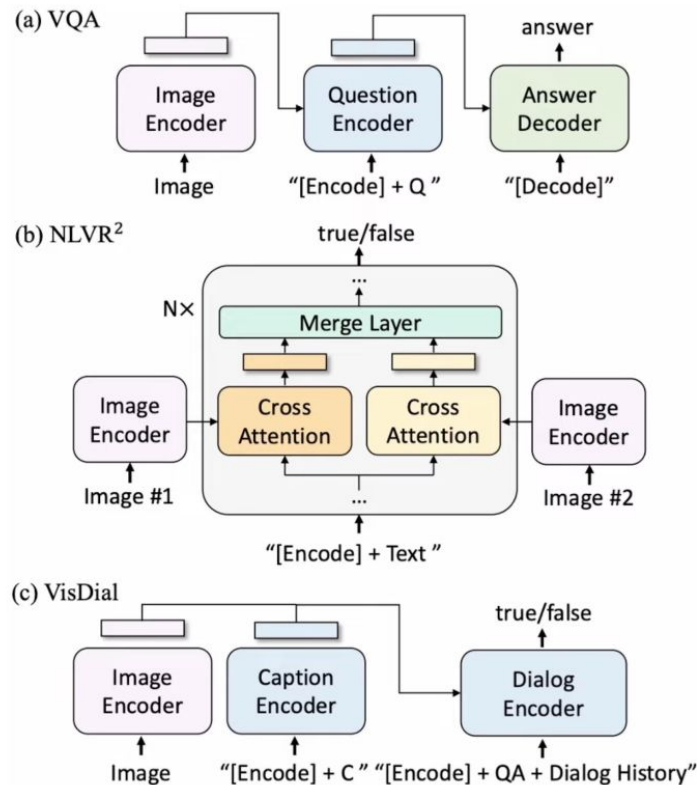
Layers with same color share weights!



Layers with same color share weights!

BLIP

- BLIP layers could be arranged to form new models suitable for specific use-cases.



VL-T5

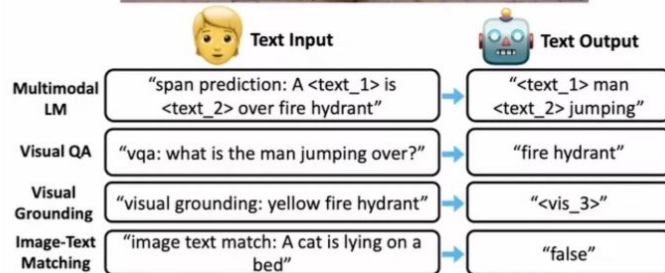
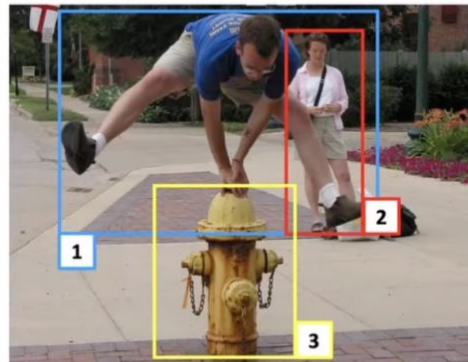
Unifying Vision-and-Language Tasks via Text Generation

Jaemin Cho¹ Jie Lei Hao Tan Mohit Bansal

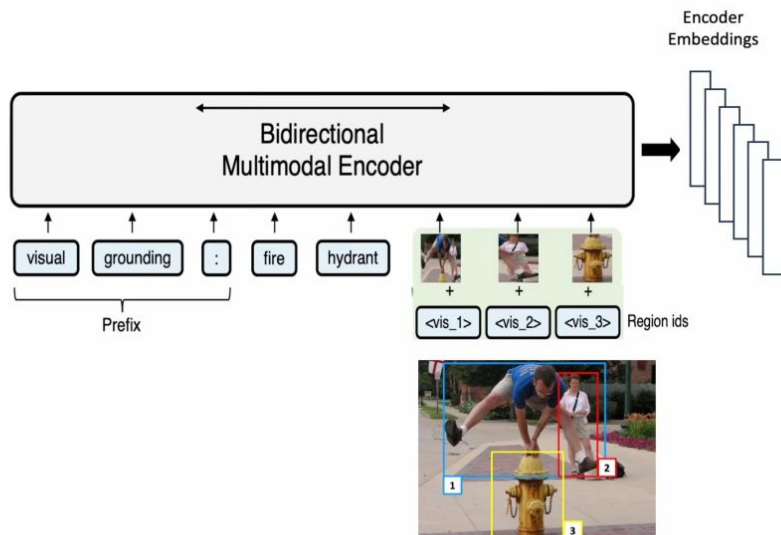
UNC Chapel Hill

{jmincho, jielei, haotan, mbansal}@cs.unc.edu

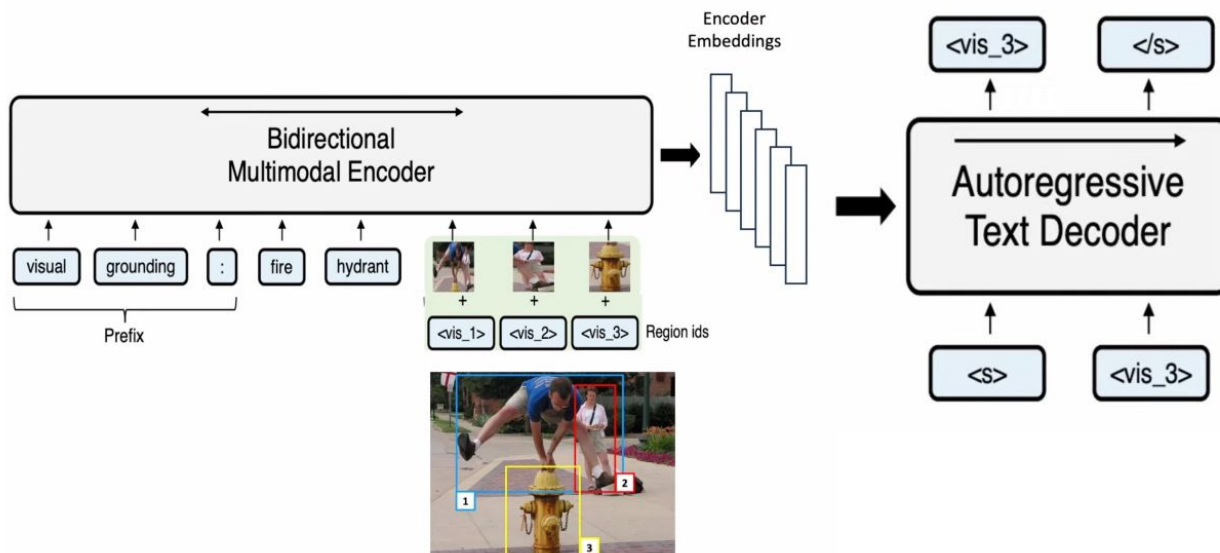
- **VL-T5** is a **multimodal encoder-decoder transformer model** for text generation



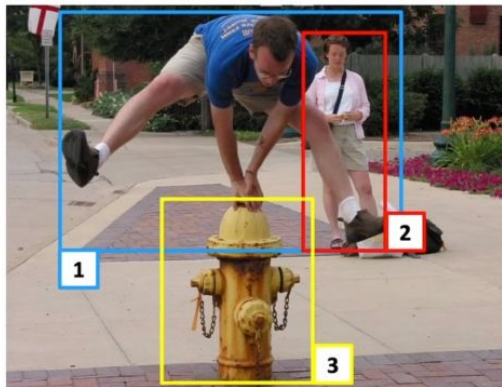
VL-T5: Multimodal Encoder-Decoder



VL-T5: Multimodal Encoder-Decoder



VL-T5 Visual Tasks



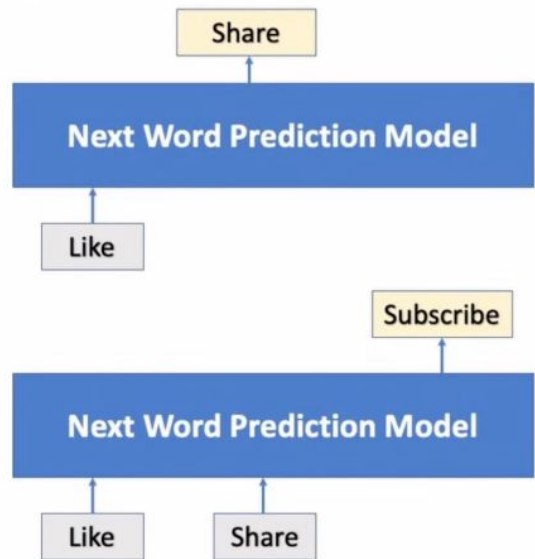
	 Text Input	 Text Output
Multimodal LM	"span prediction: A <text_1> is <text_2> over fire hydrant"	"<text_1> man <text_2> jumping"
Visual QA	"vqa: what is the man jumping over?"	"fire hydrant"
Visual Grounding	"visual grounding: yellow fire hydrant"	"<vis_3>"
Image-Text Matching	"image text match: A cat is lying on a bed"	"false"

Model can identify
image regions from
prompt!

Generative Multimodal LLMs

(Multimodal Decoder-only Transformer Models)

- **Generative Multimodal Models** use pretrained Large Language Models
 - Decoder-only Transformer architectures
 - Next Word Prediction
- DeepMind **Frozen** (2021)
- DeepMind **Flamingo** (2022)
- Meta **LLaVA** (2023)
- OpenAI **GPT-4V** (2023)
- Google **Gemini** (2023)



Fronzen (2021)

Multimodal Few-Shot Learning with Frozen Language Models

Maria Tsimpoukelli*
DeepMind
mrts@deepmind.com

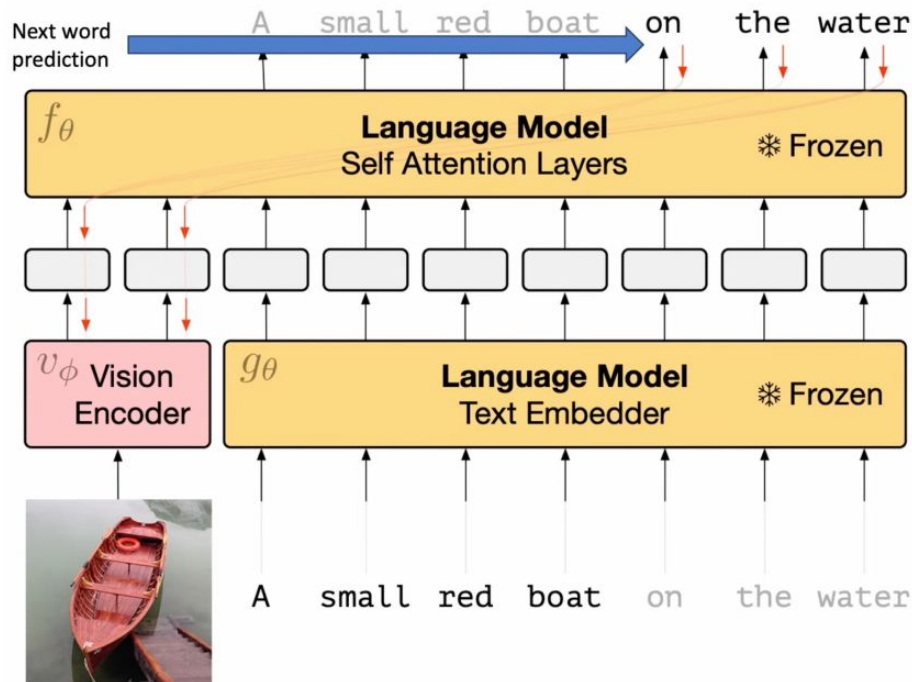
Jacob Menick*
DeepMind
University College London
jmenick@deepmind.com

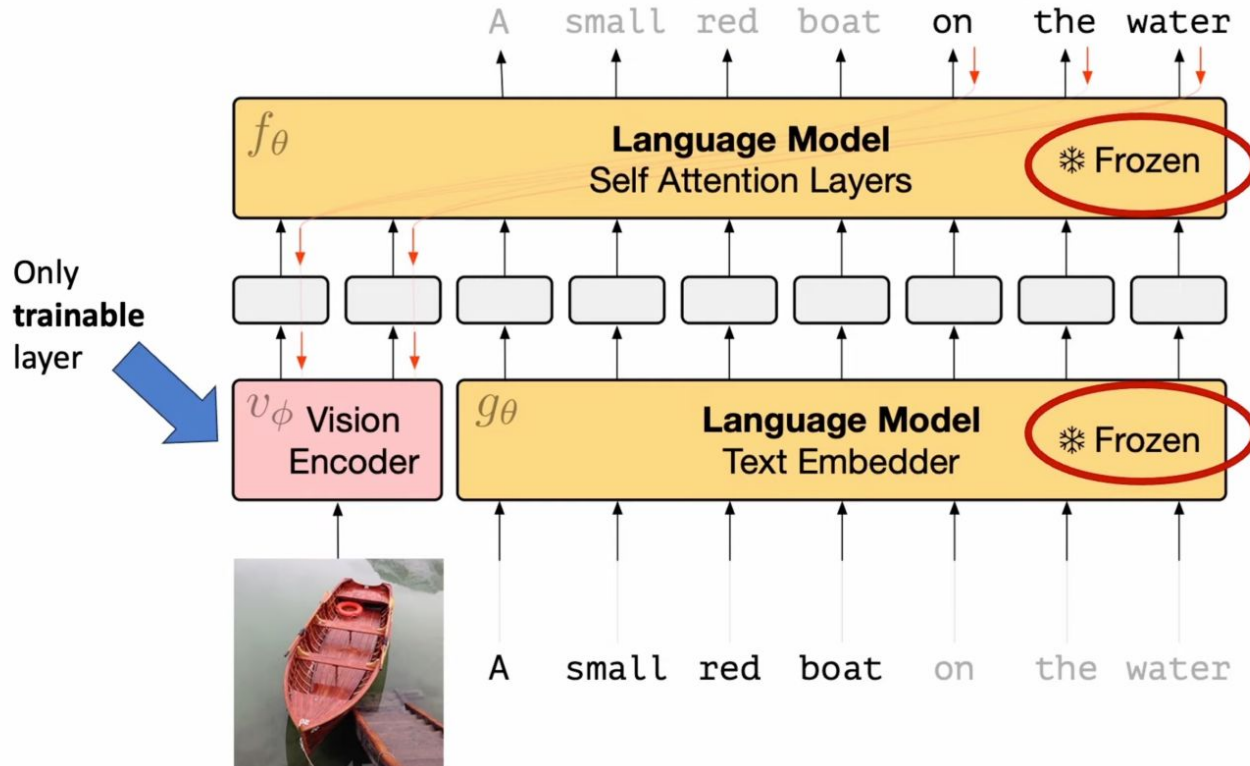
Serkan Cabi*
DeepMind
cabi@deepmind.com

S. M. Ali Eslami
DeepMind
aeslami@deepmind.com

Oriol Vinyals
DeepMind
vinyals@deepmind.com

Felix Hill
DeepMind
felixhill@deepmind.com





Context



This is a 🐎.



This is a 🐱.



This is a

Model Completion

🐶. <EOS>



This person is like 😊.



This person is like 😞.



This person is like

Model Completion

😱. <EOS>



This was invented by Zacharias Janssen.



This was invented by Thomas Edison.



This was invented by

Model Completion

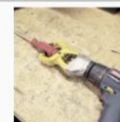
the Wright brothers. <EOS>



With one of these I can drive around a track, overtaking other cars and taking corners at speed



With one of these I can take off from a city and fly across the sky to somewhere on the other side of the world



With one of these I can

Model Completion

break into a secure building, unlock the door and walk right in <EOS>

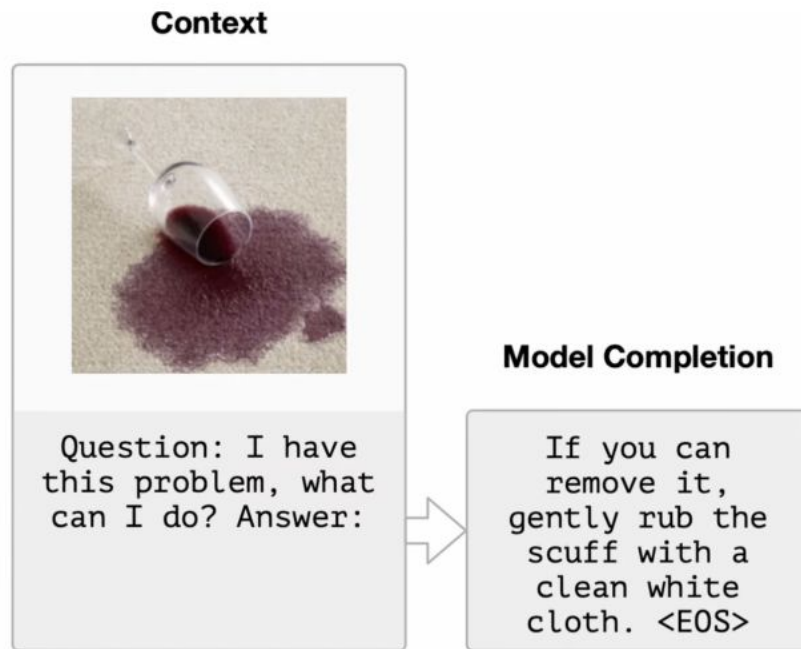


Figure 11: Encyclopedic Knowledge. Demonstrates knowledge from language pre-training being commanded given visual input. Required a few seeds to get a good answer which clearly paid attention to the image.

Context



This was invented by
Zacharias Janssen.



This was invented by
Henry Ford.



This was invented by

Model Completion

the Wright
brothers.

Figure 9: Encyclopedic Knowledge. Shows the model retrieving factual knowledge given visual input. Required cherry-picking from around 5 seeds to get past common language model failure modes like simply repeating text from the prompt or emitting text that does not pertain to the test image.

Frozen: Visual Question Answering (VQA)

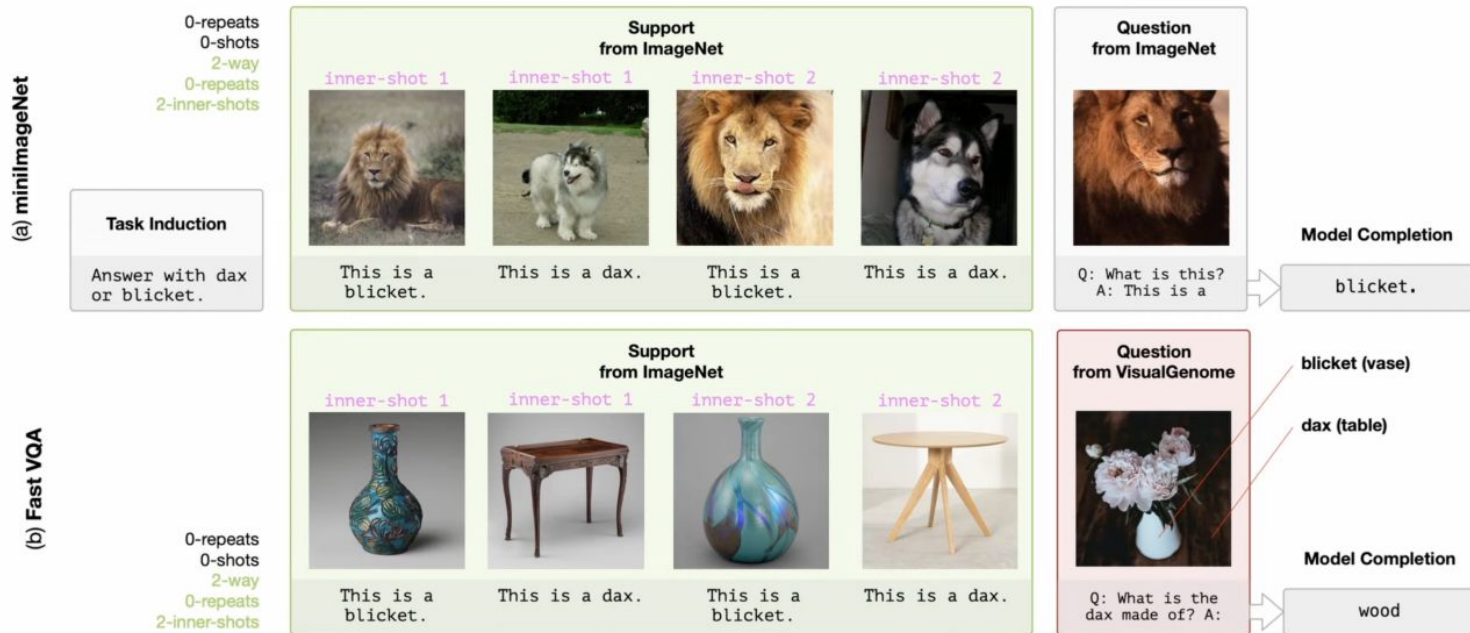


Figure 4: Examples of (a) the Open-Ended miniImageNet evaluation (b) the Fast VQA evaluation.

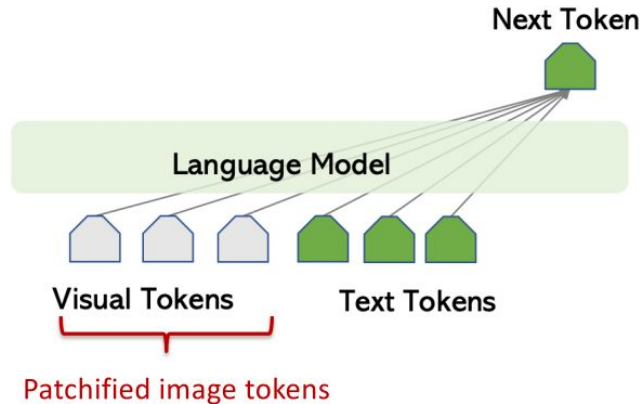
- 

Jean-Baptiste Alayrac [†]	Jeff Donahue*	Pauline Luc*	Antoine Miech*	
Iain Barr [†]	Yana Hasson [†]	Karel Lenc [†]	Arthur Mensch [†]	Katie Millican [†]
Malcolm Reynolds [†]	Roman Ring [†]	Eliza Rutherford [†]	Serkan Cabi	Tengda Han
Zhitao Gong	Sina Samangooei	Marianne Monteiro	Jacob Menick	
Sebastian Borgeaud	Andrew Brock	Aida Nematzadeh	Sahand Sharifzadeh	
Mikolaj Binkowski	Ricardo Barreira	Oriol Vinyals	Andrew Zisserman	
Karen Simonyan* [†]				

- # 🦩 Flamingo: a Visual Language Model for Few-Shot Learning

www.aiquest.org

1. **Vision encoder**: a CLIP-like model is trained by contrastive learning, which is frozen during the training.
2. **Language model**: Finetuned **Chinchilla LLM** is used to generate text tokens, conditioned on visuals and text, using language model loss, with two additional components **Perceiver Resampler** (adapter) and **GATED XATTN-DENSE layers**.



- Inject vision vectors throughout the LLM and training using long multi-image documents

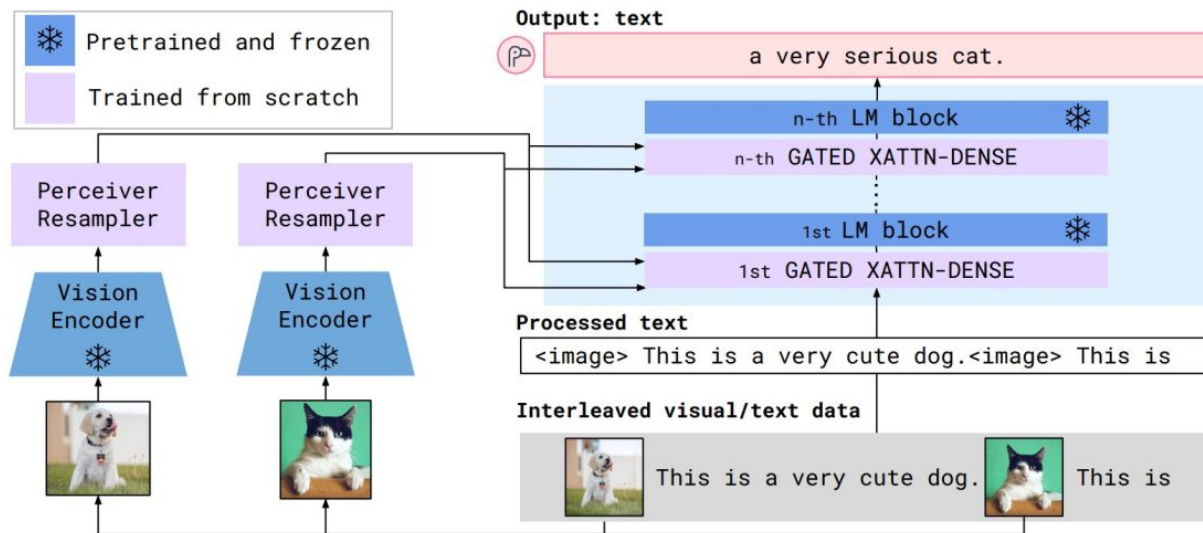


Figure 3: **Flamingo architecture overview.** Flamingo is a family of visual language models (VLMs) that take as input visual data interleaved with text and produce free-form text as output.

- **Flamingo used 4 datasets:** 2 (image, text) pair datasets, 1 (video, text) pair dataset, and 1 interleaved image and text dataset.



Image-Text Pairs dataset



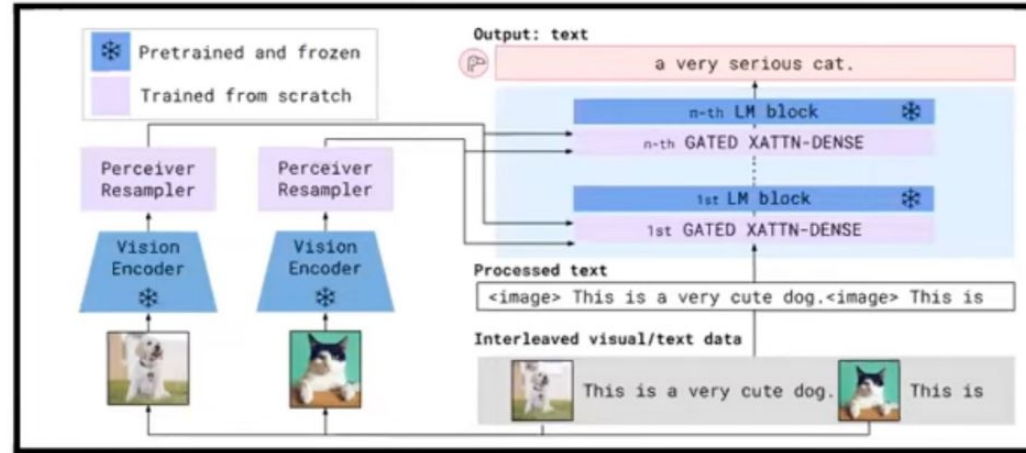
Video-Text Pairs dataset



Multi-Modal Massive Web (M3W) dataset

Dataset	Type	Size	How	Training weight
M3W	Interleaved image and text dataset	43M webpages	For each webpage, they sample a random subsequence of 256 tokens and take up to the first 5 images included in the sampled sequence.	1.0
ALIGN	(Image, text) pairs	1.8B pairs	Texts are alt-texts, averaging 12 tokens/text.	0.2
LTIP	(Image, text) pairs	312M pairs	Texts are long descriptions, averaging 20.5 tokens/text.	0.2
VTP	(Video, text) pairs	27M short videos	~22 seconds/video on average	0.03

Flamingo Train with Interleaved Image-Text Data



Pre-trained: 70B Chinchilla

Perceiver Resampler
Gated Cross-attention + Dense

Pre-trained: Nonnormalizer-Free ResNet (NFNet)

Flamingo: Zero-Shot Visual QA/Dialog

Input Prompt	Image	Question	Answer
		Question: What nutrients is the fruit in the image rich in?	Vitamin C, potassium, and fiber.
Completion		Question: What is the purpose of this?	To keep the mosquitoes out of the bed.



This is a dog. It's a white fluffy dog.

What season is this?

It looks like a summer day.

Why do you think so?

The grass is green and the dog is not wearing a sweater.

What breed is the dog?

It's a Samoyed.

Is the dog running?

No, it's sitting.

Can you describe the pose of its back legs?

The back legs are bent and the dog is sitting on its haunches.



Which city is this?

This is a street sign in New York City.

Which street corner is it?

It is the intersection of Madison Avenue and East 42nd Street.

Which subway station is nearby?

The closest subway station is Grand Central Station.

How about the nearest park?

The nearest park is Bryant Park.

There is another sign. What does it say?

It says "One way".

I would like you to read the color and then say the color it is written in. For example:

PURPLE

Color is "Purple" and it is written in red.

GREEN

Color is "Green" and it is written in blue.

YELLOW

Color is "Yellow" and it is written in green.

BLACK

Color is "Black" and it is written in yellow.

Well done! Do you know the name of the test these images come from?

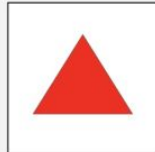
I think it is called the Stroop test.

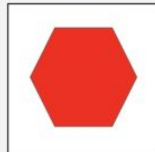
Can you explain how humans perform in this test?

Humans are slower when the color of the word and the color of the word are different.

How about you?

I am not affected by this difference.





What is the difference between these two images?

The first one is a triangle and the second one is a hexagon.

Which one has larger internal angles?

The hexagon has larger internal angles.

What color are they?

They are both red.

Flamingo: Zero-Shot Visual QA/Dialog

Input Prompt	Image	Question	Answer
		Question: What nutrients is the fruit in the image rich in?	Answer: Vitamin C, potassium, and fiber.
Completion		Question: What is the purpose of this?	To keep the mosquitoes out of the bed.



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Color is "Black" and it is written in yellow.

Well done! Do you know the name of the test these images come from?

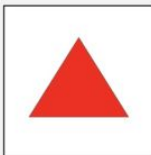
I think it is called the Stroop test.

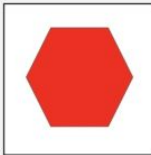
Can you explain how humans perform in this test?

Humans are slower when the color of the word and the color of the word are different.

How about you?

I am not affected by this difference.





What is the difference between these two images?

The first one is a triangle and the second one is a hexagon.

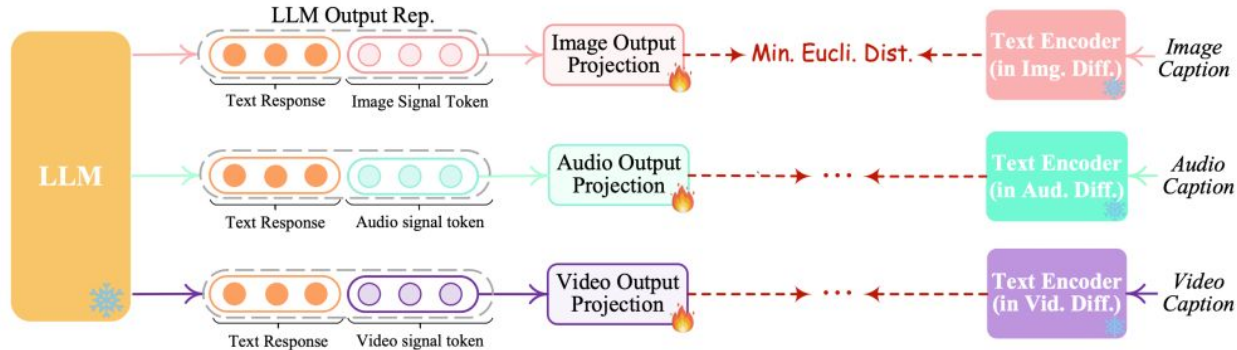
Which one has larger internal angles?

The hexagon has larger internal angles.

What color are they?

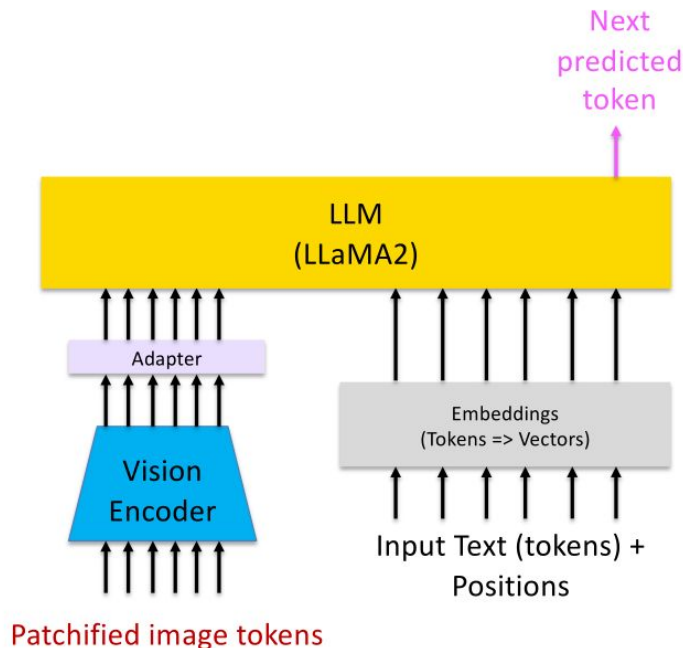
They are both red.

- **Flamingo was trained for completion, but not for dialogue or for following instructions.** Many people are working on building multimodal models that can follow instructions and have conversations.
- **LLaVA (Large Language and Vision Assistant)**
 - [LLaVA: Visual Instruction Tuning](#) (Liu et al., Apr 28, 2023) uses Decoding-side Instruction-Following Alignment



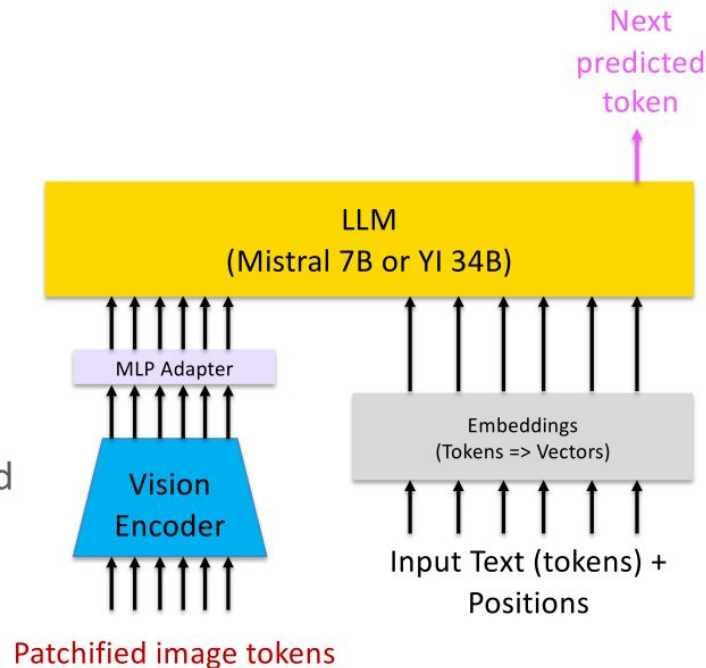
LLaVA 1.5 (2023-10)

- CLIP image encode + LLaMA2 LLM
- **Freeze** the vision encoder + LLM => **train the Adapter**
 - Use image + text datasets
- Unfreeze all parameters => Train all parameters
 - Use synthetic image + text data
 - ChatGPT (non-vision) is used to write detailed descriptions of images given details of their contents



LLaVA 1.6 (2024-01)

- CLIP encode + Mistral 7B or YI 34B LLM
- Use an MLP instead of a simple linear layer (e.g. linear layer + activation layer)
- Use a larger vision model 336x336 not 224x224.
 - ChatGPT (non-vision) is used to write detailed descriptions of images given details of their contents



GPT-4 (2023-03)



- **GPT-4** is a **large multimodal language model** created by OpenAI.
- GPT-4 has been shown to achieve human-level performance on a variety of tasks, including generating text, translating languages, answering questions, and following instructions.
- GPT-4 is still under development, but it has already been made available to the public through the paid chatbot product ChatGPT Plus, and via OpenAI's API.
- GPT-4 has the potential to revolutionize a variety of industries, and is likely to have a significant impact on society.
- It is important to use GPT-4 responsibly and ethically, and to be aware of its potential risks.

GPT-4 with Vision

Example of GPT-4 visual input:

User What is funny about this image? Describe it panel by panel.



Source: <https://www.reddit.com/r/bnmm/comments/ubab5v/bnmm/>

GPT-4 The image shows a package for a "Lightning Cable" adapter with three panels.

Panel 1: A smartphone with a VGA connector (a large, blue, 15-pin connector typically used for computer monitors) plugged into its charging port.

Panel 2: The package for the "Lightning Cable" adapter with a picture of a VGA connector on it.

Panel 3: A close-up of the VGA connector with a small Lightning connector (used for charging iPhones and other Apple devices) at the end.

The humor in this image comes from the absurdity of plugging a large, outdated VGA connector into a small, modern smartphone charging port.

GPT-4 visual input example, Extreme Ironing:

User What is unusual about this image?



Source: <https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg>

GPT-4 The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.

Table 16. Example prompt demonstrating GPT-4's visual input capability. The prompt requires image understanding.

GPT-4V

- GPT-4V generates Latex code, which can then be reconstructed as a data table

Prompt:

Please read the table in this image and return a latex-style reconstructed table in text.

Country	Sales Volume	Revenue	Profit
USA	40.080	\$15,971.880	\$3,086.421
China	35.070	\$15,866.670	\$3,032.162
Australia	27.054	\$14,812.566	\$2,868.636
India	23.046	\$10,608.174	\$1,853.710
South Korea	16.032	\$10,494.948	\$1,975.844

Note: Same query image as shown above, resized to save space

GPT-4V:

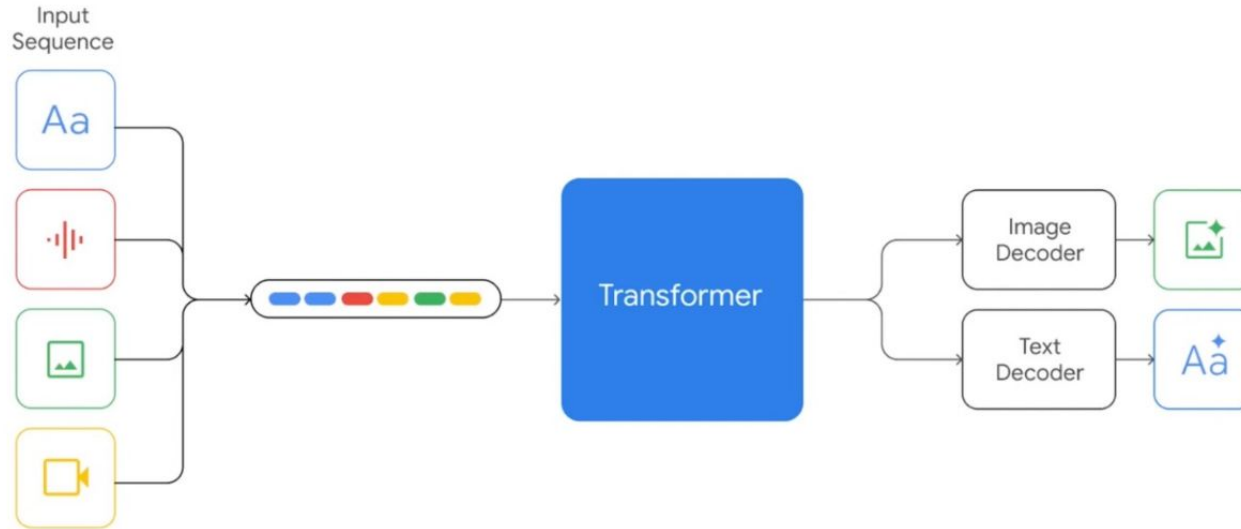
Here is the reconstructed table in latex:

```
\begin{tabular}{|c|c|c|c|} \hline
Country & Sales Volume & Revenue & Profit \\ \hline
USA & 40.080 & $15,971.887 & $3,068.421 \\ \hline
China & 35.070 & $15,866.670 & $3,032.162 \\ \hline
Australia & 27.054 & $14,812.566 & $2,868.636 \\ \hline
India & 23.046 & $10,608.174 & $1,853.710 \\ \hline
South Korea & 16.032 & $10,494.948 & $1,975.844 \\ \hline
\end{tabular}
```

Visualization of Reconstructed Table (LaTeX):

Country	Sales Volume	Revenue	Profit
USA	40.080	\$15,971.887	\$3,068.421
China	35.070	\$15,866.670	\$3,032.162
Australia	27.054	\$14,812.566	\$2,868.636
India	23.046	\$10,608.174	\$1,853.710
South Korea	16.032	\$10,494.948	\$1,975.844

Google Gemini (2023)





<https://www.youtube.com/watch?v=UIZAiXYceBI>



OpenAI just introduced Sora, their new text to video AI model, able to produce up to 60 seconds of high-quality video, blowing the competition out of the water.

https://www.youtube.com/watch?v=nFGPoJ7_OTY